Your grade: 90%

Your latest: **90%** • Your highest: **90%** • To pass you need at least 70%. We keep your highest score.

Next item →

1. In how many ways can 3 students be selected from a group of 5 to stand in line? 1 / 1 point

- 20
- 60
- 30
- 10

Nice work

The answer is $P(5,3) = 60$. You can also solve it by calculating $5 \times 4 \times 3 = 20$

2. Consider the following two sets $A = \{1,2,3,4\}$ $B = \{a,b,c,d,e\}$. In how many ways we can create a 4-character password such that it contains two elements from A and two elements from B? 1 / 1 point

- 240
- 1440
- 480
- 960

Nice work

We have $\binom{5}{2}$ ways to pick two elements from B and $\binom{4}{2}$ ways to pick two elements from A. They also can form 4! permutations. So the answer is $\binom{5}{2} \times \binom{4}{2} \times 4!$.

3. 5 countries participate in a competition by each sending 10 athletes. In how many ways we can pick 3 athletes such that they all belong to different countries. 1 / 1 point

- 10^3
- 10^4
- 10^6
- 10^5

Nice work

Note that we first must pick 3 countries from 5 which can be done in $\binom{5}{3}$ different ways. Now from each country, there are 10 ways to pick an athlete. So the answer is $\binom{5}{3} \times 10^3 = 10^4$.

4. Assume the CEO, CTO and 4 employees are planning to sit around a round table. In how many ways the CEO and CTO are sitting exactly in front of each other. 1 / 1 point

- 40
- 60
- 48
- 24

Nice work

We know that these 6 people have $5!$ ways to sit around a table. Consider the CEO to be seated anywhere. The CTO, has 5 places to sit, however in only one of them he is in front of the CEO. So the answer is $\frac{5!}{5} = 24$

5. Assume 5 people $\{a,b,c,d,e\}$ are sitting next to each other in a line. In how many ways a and b are sitting next to each other? 1 / 1 point

- $5! \cdot 2!$
- $\frac{5!}{2!}$
- $5!$
- $4! \cdot 2!$

Nice work

We can assume a, b are attached to each other and form a ``pack". They have $2!$ permutations among themselves. And now there are 3 other people and their ``pack" and they have 4! Permutations. So the answer is $4! \times 2!$.

6. How many non-negative solutions exist for $x_1 + x_2 + x_3 = 10$ such that $x_1 > 3$? 0 / 1 point

- $\binom{12}{2}$
- $\binom{8}{2}$
- $\binom{12}{2} - \binom{11}{2}$
- $\binom{9}{2}$

Try again

7. Consider the following set $\{1,2,\cdots,10\}$. In how many ways can we pick 4 numbers such that two of them are even and two of them are odd. 1 / 1 point

- 80
- 100
- 112
- 120

Nice work

There are 5 odd numbers and 5 even numbers. The answer is $\binom{5}{2} \times \binom{5}{2} = 10 \times 10 = 100$

8. Assume $\binom{n}{8} = 2\binom{n}{7}$. What is the value of n? 1 / 1 point

23

Nice work

9. How many ways can we arrange the letters in the word ``SUCCESS"? 1 / 1 point

- $\frac{7!}{2! \times 2!}$
- $7!$
- $\frac{7!}{2! \times 3!}$
- $\frac{7!}{2}$

Nice work

There are $7!$ permutations in total. However, as there are two ``C" and three ``S", each case appears $2! \times 3!$ times. So we must divide $7!$ by $2! \times 3!$.

10. Consider the previous question, but now there are only 3 different pop songs and 4 different classical songs. Assume you want to listen to them in an alternate way. In how many ways can you do that? 1 / 1 point

- $3! \times 3!$
- $2 \times 3! \times 4!$
- $3! \times 4!$
- $2 \times 3! \times 3!$

Nice work

Note that now the order can only be $CPCPCPC$. So the answer is $3! \times 4!$.